MSc TP Curriculum for 2024 and 2025 intakes

Minimum Credit Requirement

48 credits
Core Courses: 21
Elective Courses: 27

* Subject to the approval of the Program Director, students may apply for credit transfer or course substitution of no more than 12 credits.

1 Core courses

Course code	Course title	Credits	Course description
IPEN 5110	Foundation in Public Policy	3 credits	The course will provide an advanced foundation in the study and practice of public policy at the level required for graduate study. The course will cover both the historic foundations of policy studies, as well as emerging approaches and directions. As the study of public policy is inherently interdisciplinary, it will include perspectives from political science, public policy, economics, business and other aspects of social science. It will take a broad view of public policy, including taking up some of the core literature on public management and public administration.
IPEN 5130	Economics of Technology Innovation and Entrepreneurship	3 credits	This course introduces the economics of technology innovation and entrepreneurship through the combined perspectives of microeconomics and macroeconomics. It covers microeconomic core modules concerning consumers, firms, markets, and governments, as well as macroeconomic core modules on economic growth associated with entrepreneurship and innovation.
IPEN 5150	Policy Analysis for Technology and Innovation	3 credits	Technological innovation is increasingly the source of sustainable competitive advantage for firms worldwide. This course introduces a grounding in the field of technology and innovation, with an emphasis on economic policy and business strategy. The course will be highly interactive and apply multiple disciplines including economics, management, law and public policy.
IPEN 5160	Big Data Applications for Business and Government	3 credits	This course will cover the key concepts and technologies of big data and data analysis, with a focus on the application of big data in formulating business strategies and policies, and related research issues on how big data affects the direction of business and policy development. The course will provide students with practical training on big data and data analysis based on real-world business or policy issues, ranging from collecting and preprocessing to organizing and analyzing large-scale data.
IPEN 5180	Disruptive Technology and Society	3 credits	This course gives students a broad introduction to the key disruptive technologies, such as mobile internet, AI, and robotics, that have transformed our society. We will examine the practical applications of these technologies and discuss

			their socioeconomic impacts and policy responses. We will also look at the potential for businesses and governments to harness these disruptive technologies to deliver new services or improve existing ones and enhance value in public and private sectors.
IPEN 6110	Capstone Project	6 credits	This course consists of 6 credits and will last for two regular terms. In the first term, students learn and integrate the latest technology topics through seminars, lectures, and workshops. After completing four micro-policy analysis reports, students can familiarize themselves with technology policy hot spots and policy analysis tools, as well as the cooperation skills and role division among the groups. In the second term, they will complete group projects on selected topics for science and technology policy under the supervision offaculty members. The participation of the university's internal community and external organizations in these projects will be highly encouraged. The university will be responsible for the control, management and evaluation of the project. Students will exercise their teamwork skills, analyze science and technology policy issues and develop concise reports of their findings and recommendations. They should write the paper acting as an assistant to a particular decision-maker in a government, nonprofit organization, business or private sector. This course is for MSc(TP) students only. May be graded PP.

2 Elective Courses

To meet the elective course requirements and individual needs, students can take the courses in different areas, which may include but not limited to courses and areas listed below.

Additional coursework may be required as part of the program preparation.

Course code	Course title	Credits	Course description
IPEN 5111	Public Management and Institutional Analysis	3 Credits	This course focuses on the theoretical and analytical perspective of public management and institutions. It introduces students to key concepts in the discipline of public management and institutional analysis. The course begins with a review of the evolution of thinking in this field. In the following sessions, students will be extensively exposed to theoretical frameworks. The course aims to equip students with theories that help students in building up their capacity toward academic research.
IPEN 5120	Research Design for Innovation, Policy and Entrepreneurship Studies	3 Credits	The purposes of the course are to introduce to students key concepts in research design, and to help them develop skills in the design of empirical research for conducting innovation, policy and entrepreneurship studies. Specific emphasis will be on the use of quasi-experimental designs in policy research, as well as on their potentials and limitations.

IPEN 5140	Quantitative Analysis and Empirical Methods	3 Credits	This course introduces students to empirical methods and data management tools used in the current social science disciplines, with some special focuses on strategy, finance and applied micro-economics. The overall approach is to understand the common methods and research design used in the empirical research through intensive reading and replicating papers published in top journals. Students would also become proficient in the use of computer software that is widely used in analyzing quantitative data via empirical assignments.
IPEN 5200	Uncertainty, Information and Decision Making	3 Credits	This course introduces the economic theories of decision making under risk and uncertainty and how agents with heterogeneous information interact strategically. Sample topics include expected and non-expected utility theories, models of strategic communication, and information design. Students will apply the theoretical tools to understand and improve real world institutions, such as employee feedback systems and transparency in organizations.
IPEN 5250	Text Analysis and Machine Learning	3 Credits	This course serves as an applied introduction to machine learning methods for text analysis. Several approaches on text data management and analysis will be covered in this course including basic natural language processing techniques, document representation, text categorization and clustering, document summarization, sentiment analysis, social network and social media analysis, probabilistic topic models and text visualization.
IPEN 5260	Corporate Governance Research	3 Credits	This course is designed to introduce students to the key corporate governance phenomena examined in the field of strategic management. We will review how economic and organizational theories are applied to explain the choices and outcomes of governance design. In addition, we will also investigate some behavioral or process-related factors affecting the functioning of governance mechanisms, especially the board of directors.
IPEN 5270	Corporate ESG Practice and Research	3 Credits	This course focuses on the development of the ESG (Environmental, Social and Governance) movement and its relationship with other long-standing concepts such as corporate social responsibility and sustainability. It aims to review the content, antecedences, and consequences of corporate ESG practices and stimulate new research ideas in related areas.
IPEN 5280	Technological Catching-up Policies and Management	3 Credits	This course deals with various issues when late-comer firms and countries may encounter in the technological catching-up with the more advanced firms and countries. Technological stages and paths are studied and highlighted at the three different levels of country, sector and firms. Students will learn various theories regarding technological catching-up and have opportunities to apply them to real cases.
IPEN 5290	Public Organizations and Policy Design	3 Credits	This course is designed to provide learning opportunities regarding how to analyze and structure messy unstructured public problems. This course consists of a series of different fake public problem cases to help students experience setting up, analyzing, and designing policies. Over the course of the term, students analyze the cases quantitatively and qualitatively, and then propose solutions with an integrated manner of the analysis results. Much of the work is done in small groups or individually.
IPEN 5300	Experimental Economics and	3 Credits	This course introduces the methodology of experimental economics and related behavioral theories, with an emphasis on social-psychological

	Organizational Behavior		elements of preference and organizational design. Experiments studied will include ones based on the prisoners' dilemma, dictator game, ultimatum game, and especially the public goods game and the trust game, along with more complex designs for studying institutional and organizational problems such as creation of centralized punishment schemes and secure property.
IPEN 5310	Behavioral Economics and Public Policy	3 Credits	This course introduces behavioral economics - the incorporation of insights from psychology into economics - with an emphasis on its value for improving empirical predictions and policy decisions. Students will learn the major themes of behavioral economics and apply them to improve the design, implementation, and evaluation of public policies in a wide variety of domains.
IPEN 5330	Energy Economics and Policy	3 Credits	This course provides an in-depth examination of the economics and policy issues surrounding energy. Topics include energy demand and supply, market structures, renewable energy technologies, fossil fuels, energy security, climate change, transportation, innovation, and policy analysis. Students will also participate in a research project to apply their knowledge to real-world scenarios.
IPEN 5340	Business Model Innovation for Sustainability	3 Credits	This course provides an overview of sustainable business model innovation theories and practices as well as discusses how business models can serve as essential tools to transform our economy into a sustainable one. We will provide students with a rich opportunity to examine companies' efforts in creating and implementing sustainable business models and building a supporting ecosystem.
IPEN 5350	Energy Systems and Policy	3 Credits	This course explores the challenges associated with climate change, energy and environmental policies from multiple perspectives, disciplines and scales. Students will examine the evolving science and policy of climate change, observe the conflicts between energy companies and interest groups in China, discuss technology innovations and its impact on reducing greenhouse gases emission.
IPEN 5400	Climate Change: Science and Governance	3 Credits	This course prepares students to acquire the basic knowledge of climate change, which sits on the intersection of science and governance. It will review some of the scientific facts of climate change and contrast the scientific research findings with climate governance status. Case study on transforming to a low carbon society will be conducted in later part of the course. Aspects to consider include both scientific support and governance complexity of the low carbon city idea. Students are expected to build their own analysis of the climate change issue at the end of the course.
IPEN 5700	Strategic Management of Technology and Innovation	3 Credits	Technological innovation is increasingly the source of sustainable competitive advantage for firms worldwide. This course introduces a grounding in the field of technology and innovation, with an emphasis on economic policy and business strategy. The course will be highly interactive and apply multiple disciplines including economics, management, law and public policy.
IPEN 5810	Data Science in Empirical Economics	3 Credits	In the digital age, there is more data available than ever before on human behavior: from analyzing an elected official's opinion on Twitter to identifying a farmer's crop choices through satellite images. This course aims to familiarize students in applied economics, public policy, and relevant disciplines with recent research that has used big data to push

			the cutting-edge of the applied economic and public policy fields. Through a combination of problem sets and independent projects, students will acquire the statistical and computational tools needed for making use of big data in empirical research.
FTEC 5030	Statistical Methods for Financial Technology	3 Credits	This course will survey modern financial technology, through the lens of statistics, which is the science of the analysis of data. Students will learn how statistical methodology, in conjunction with advances in technology, is used to efficiently acquire, utilize and interpret data, as it relates to innovations in the financial services sector. This course will develop skillsets for Big Data analytics and Predictive modelling, for better understanding of the financial markets.
FTEC 5031	Advanced Probability Theory	3 Credits	The course will give students a deeper understanding of the foundations of probability theory, such as probability theory from a measure-theoretic perspective, convergences of distributions and probability measures, and conditional expectations. During the course, important theorems, such as Radon-Nikodym theorem, Fubini theorem, and general central limit theorems, will be investigated.
FTEC 5050	Machine Learning and Artificial Intelligence	3 Credits	This course covers the fundamentals of machine learning and artificial intelligence, and their applications in computer vision, image processing, natural language processing, and robotics. The topics include major learning paradigms (supervised learning, unsupervised learning and reinforcement learning), learning models (such as neural networks, Bayesian classification, clustering, kernels, feature extraction), and other problem solving techniques (such as heuristic search, constraint satisfaction solvers and knowledge-based systems) in AI.
FTEC 5100	Research in Corporate Finance	3 Credits	This course introduces the main issues in corporate finance, identifies principal theoretical tools and empirical approaches, and fosters thinking about current research questions. The theoretical part includes classic theories such as Modigliani-Miller theorem, Coase theorem, and Fisher separation theorem, with a focus on financing decisions of firms, corporate governance, and their implications. The empirical part reviews econometric methods commonly used in corporate finance research and covers selected topics.
FTEC 5101	Microeconomic Theory	3 Credits	This is a course in graduate level microeconomic theory for PhD students in financial technology and other related fields. This course covers topics including consumer theory, producer theory, uncertainty, general equilibrium,and matching. The required background knowledge for the course are intermediate microeconomic theory and mathematics through calculus of several variables and introductory real analysis. Additional mathematical tools will be explained briefly as the course proceeds. This course serves as the first rigorous training in economics and finance and helps lay down a solid foundation in economic modelling for future research.
FTEC 5210	Quantitative Models for Financial Derivatives (3 units)	3 Credits	This course covers basic pricing theory of financial derivatives and risk hedging of exotic options. The course starts with the fundamental theorem of asset pricing and risk neutral valuation principle. The renowned Black-Scholes pricing theory and martingale pricing theory are introduced. Advanced topics include exchange options, quanto options, implied volatility and VIX.

FTEC 5230	Financial Risk Management	3 Credits	This course shows the use of various quantitative techniques and financial engineering principles in the management and modeling of financial risks. The topics include hedging of market risks, immunization of bond risks, Value-at-Risk and expected shortfall, credit yield curve model, credit derivatives pricing, and default correlation models.
FTEC 5310	Blockchain Technology	3 Credits	This course offers a thorough introduction of blockchain technology. Students will explore both technical and non-technical aspects of blockchain, including but not limited to cryptographic techniques, consensus mechanisms, smart contracts, regulatory implications, and real-world applications. Topics covered in this course include the historical development of blockchain, Bitcoin protocol, Ethereum, applications of blockchain, such as decentralized finance, and regulation discussions.
UGOD 5009	Inferential Statistics and OLS Regression	3 Credits	This course introduces fundamental knowledge and practice of basic statistics in quantitative social science research, with a focus on how quantitative methods are used to assemble, describe, and draw inferences from bodies of numerical data. The course serves as an additional foundation for more advanced methodology courses (such as UGOD 5020). The course covers two modules. The first is about descriptive statistics and fundamentals of statistical inference. Topics include frequency distribution, probability theory, random variable and probability distributions, estimation, hypothesis testing, t-test, Analysis of Variance (ANOVA), and contingency table analysis. The second is about linear regression techniques, which are widely used in social science research. The course materials are explored through the analyses of real data sets using STATA.
UGOD 5020	Quantitative Social Science	3 Credits	This course builds on the knowledge of the linear regression models to introduce students advanced statistical methods to analyze survey, administrative and other types of data of interest to quantitative social scientists. The introduction of statistical methods is integrated into research contexts and designs from a holistic framework and bridge quantitative social science and computational social science (data science). Topics include measurement, prediction, causal inference, natural experiment and program evaluation (difference-in-differences, panel data, instrumental variables, regression discontinuity), applied to both survey and big data.
UGOD 5040	Urban Data Acquisition and Analysis	3 Credits	The course introduces students to different methods of collecting data in the social sciences for urban analysis, focusing on sampling surveys designs and analysis in urban settings. Since alternative data sources (e.g., passive measurement, social media and administrative data) become increasingly available in recent years, the course will also cover other modes of data acquisitions such as using new technology on wearables, sensors, and apps in urban research settings, and exploration of cutting edge methods for collecting and analyzing web data, and how they can be used in combination with traditional survey data.
UGOD 5050	Cities and Society	3 Credits	The course looks at some of the major drivers of urban inequality and poverty, and the key actions that cities are taking to reduce urban inequalities through urban design, infrastructure and policy. Students are introduced with tools to analyze the socio-demographic profile of households and neighborhoods/communities and their relation to spatial

			distribution and clustering in cities of both the developing and the developed world. A particular emphasis is placed on identifying spatial strategies that can alleviate the concentration of urban poverty and inequality to enhance urban social cohesion by optimizing access to jobs, housing, education, health, public space, transport and community infrastructure.
UGOD 5070	Urban Planning and Design	3 Credits	This course cuts across all major fields within urban planning and design and introduces the major theories, models, and methodological approaches that urban planners and policy makers use for urban planning and design. This course also critically examines the current practice of urban planning and governance in China at various geographical scales.
UGOD 5080	Economic Analysis of Cities and the Environment	3 Credits	This interdisciplinary course will provide students a systematic framework of the interplay between urban growth and the environment from economic perspectives.By walking them through the state-of-the-art research in urban and environmental economic studies from both developing and developed countries., it will familiarize students with popular empirical strategies in applied economics and relevant fields to solve the most pressing environmental challenges accompanied with fast-urbanized cities. By the end of the class, students will be equipped with toolkits to evaluate policy questions in transportation, pollution and health, climate change, energy transition, housing market, and environmental justice.
UGOD 5101	Empirical Methods for Urban Research	3 Credits	The course introduces students to basic practices and tools that will enhance their ability to conduct empirical research and analysis in applied economics and relevant disciplines in a data-rich world. By the end of the course, the students will be proficient in a variety of data management, visualization, and quantitative techniques necessary to efficiently conduct independent research. The course format is “hands-on”, and students will conduct most of their work on their personal computers using R and RStudio.
UGOD 5105	Urban Complexity Analysis and Modeling	3 Credits	Going beyond conventional GIS and geospatial analysis, this course delves into the realm of complexity-science methods, including fractal geometry, power law statistics, space syntax, complex networks, scaling hierarchy, and cellular automata. By embracing pivotal concepts like "natural streets" and "natural cities," alongside the tools such as Axwoman and head/tail breaks, students will explore city structure and dynamics from a complexity-science perspective. These concepts, methods and tools will be applied to open-access geospatial big data such as OpenStreetMap, nighttime imagery, and location-based social media data for revealing insights into cities for better transforming modern cities towards a sustainable planet.
UGOD 5110	Quantitative Macroeconomics	3 Credits	This course introduces the framework of modern macroeconomics and then conducts policy analysis based on it. We start from the classic economic growth model developed by Solow, then gradually add other features to the model to make it better explain the real world. These features include a life-cycle structure, income risks, incomplete credit markets, the role of government, etc. After building up the framework, we study its policy implications with a focus on fiscal policy.

UGOD 5120	Entrepreneurship and Modern Economy	3 Credits	This course studies the role of entrepreneurship in the modern economy. Theories and empirics of startups as a driver of job creation, growth and development, and business cycles will be discussed. The course also examines the effects on venture growth, wealth accumulation, inequality, and innovation in a society. Finally, examples of entrepreneurial public policies will be demonstrated and assessed.
UGOD 5700	Categorical Data Analysis	3 Credits	This course focuses on statistical analysis of data that are categorical or non-continuous in nature. The contents cover a family of statistical models that deal with binary data, discrete data, count data, and categorical data with special features such as truncation and overdispersion. The course gives emphasis to modeling techniques and hands-on applications to empirical research.

The end